SIMILARITY OF FINITE CENTRAL SIMPLE ALGEBRAS AND THE ASSOCIATED SKEW FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction is stated with its reason and the remaining assertion is proved by exhibiting an invariant, so no step is left to the reader.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Wedderburn's theorem. Same chapter, Theorem theorem-wedderburn. Supplies that every finite central simple algebra is a matrix algebra over a finite central skew field, which lets the author reduce to the third assertion.
- The endomorphism ring of a simple module. Same chapter, Lemma lemma-simple-module-unique. Supplies that the endomorphism ring of a simple module over a matrix algebra is the opposite of the skew field, which distinguishes the skew fields up to isomorphism.

Source reference. Original theorem 0747: sources/stacks-brauer/pinned-original-source.tex, source lines 358–358. Navigation pointer invoked at proof line 622; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 074E: sources/stacks-brauer/pinned-original-source.tex, source lines 498–498. Navigation pointer invoked at proof line 629; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Similarity.*

Date: stacks-project:brauer.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *Similarity defines an equivalence relation on the set of isomorphism classes of finite central simple algebras over k.*
(2) *Every similarity class contains a unique (up to isomorphism) finite central skew field extension of k.*
(3) *If $A = Mat(n \times n, K)$ and $B = Mat(m \times m, K')$ for some finite central skew fields K, K' over k then A and B are similar if and only if $K \cong K'$ as k-algebras.*

Proof. Note that by Wedderburn's theorem (Theorem 0747) we can always write a finite central simple algebra as a matrix algebra over a finite central skew field. Hence it suffices to prove the third assertion. To see this it suffices to show that if $A = \mathrm{Mat}(n \times n, K) \cong \mathrm{Mat}(m \times m, K') = B$ then $K \cong K'$. To see this note that for a simple module M of A we have $\mathrm{End}_A(M) = K^{op}$, see Lemma 074E. Hence $A \cong B$ implies $K^{op} \cong (K')^{op}$ and we win. □